A hierarchical face-scalp-cortex mapping for non-invasive neuromodulation targeting without magnetic resonance images guidance

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Abstract

Neuronavigation enable precise targeting for non-invasive neuromodulation, yet remains dependent on individual magnetic resonance images (MRI) to link the external surfaces to the internal brains. Here we present a unified framework for a hierarchical face-scalp-cortex mapping that enables transcranial targeting without MRI and elucidates the feasibility of surface-to-brain spatial estimation. By uncovering the latent correlation between face and scalp topologies to establish a face-to-scalp mapping, and by statistically analyzing the coordinates of cortex targets in a new scalp angle coordinate system to construct a cortex-to-scalp mapping, we provide a principled explanation for using surface landmarks to estimate transcranial and intracranial targets. We further conducted experiments on phantoms and insomnia patients using a markerless robotic neuronavigation system, demonstrating the potential of the framework to lower clinical entry barriers and its broad clinical application prospects.

Main text

Introduction

Non-invasive neuromodulation techniques, including transcranial magnetic stimulation (TMS), transcranial ultrasound stimulation (TUS), and transcranial photobiomodulation (tPBM)[1-4], have demonstrated significant potential in the treatment of psychiatric and neurological disorders, including sleep disorders, depression, anxiety, and chronic pain[5-8]. By providing a non-pharmacological therapeutic option, non-invasive neuromodulation techniques help reduce the risk of medication overuse and associated side effects. The evolution of neuronavigation has further strengthened the precision of non-invasive neuromodulation techniques by enabling accurate targeting of specific brain regions[9,10]. Compared with traditional manual localization methods[11-13], neuronavigation techniques offer significantly greater accuracy and reproducibility of applicator placement[10,14-18]. Despite these advantages, current neuronavigation systems involve complex operational procedures[10,16,18] and can only be performed by well-trained physicians, thereby increasing their clinical workload. Moreover, precise localization depends on structural MRI as the basis, which inevitably subjects patients to lengthy examination procedures and delaying their neuromodulation treatment schedule. The high cost of MRI scanning also makes it inaccessible to many medical institutions, and some patients cannot tolerate the noise and enclosed environment of the scanning process[10,19,20]. Given these challenges, manual localization remains the pragmatic preference in treating conditions such as sleep disorders and mild depression, indicating the dilemma of balancing navigational precision and clinical convenience.

To address the inconvenience usage of neuronavigation in clinical settings, continuous studies have attempted to estimate the placement of the applicator without MRI by either providing the spatial information required by navigation software via geometric surface reconstruction or by establishing direct correspondences via surface-to-brain mapping. The former methods employ sparse sampling of scalp point locations[21-23] followed by fitting the cranial surface in a bid to reduce the reliance of neuronavigation on MRI. However, such approaches involve labor-intensive sampling procedures that require professional operation. Moreover, the geometric construction of cranial surface is highly dependent on assumptions about the topological relationships among sampled points and subsequent geometric interpolation, which poses significant limitations in practical applications. In parallel with these methods, surface-to-brain mapping aims to establish a direct link between surface landmarks and brain targets across different mapping domains. A domain with broad clinical validation focuses on cortex-to-scalp mapping, from the 5 cm method to discrete electrode-position estimation systems based on 10-20 EEG caps and their variants[12,13], and further to transcranial brain atlases constructed under continuous proportional coordinate (CPC) systems[20, 24]. The heuristic scalp approaches such as the 5cm method and the 10-20 system can approximate specific targets within Montreal Neurological Institute (MNI) brain space[11-13], but their applicability is limited to a finite number of group targets, and accuracy depends on the operator's experience. The introduction of CPC[20,24] providing a systematic modeling approach to derive cortical mapping from the scalp surface. Nevertheless, the efficacy of these methods still depends on the manual acquisition of scalp spatial data. Additionally, new domain mapping methods have emerged that use facial landmarks as predictors of brain targets or neurocranial landmarks[25-27], motivated by the potential of computer vision to reduce manual errors and bridge the expertise gap among

physicians. These methods have yielded promising results, but they lack a principled demonstration of the underlying reasoning behind these mappings.

In this paper, we propose a unified framework that breaks down the isolation between geometric surface reconstruction and surface-to-brain mapping through a hierarchical face-scalp-cortex mapping. Leveraging 3D generative artificial intelligence[28-33], the framework uncovers the latent topological correlations between face and scalp to achieve end-to-end cranial surface reconstruction. In implementation, the facial point clouds acquired by the scanner are completed into a full cranial surface model via a generative modeling, thereby eliminating the need for manual scalp measurements. Concurrently, we present a novel cortex-to-scalp mapping approach designed to adapt to the completed cranial surface. By combining spatial information from the cranial surface model, this mapping method estimates the targeting poses of MNI brain space targets on the individual's head. The hierarchical mapping framework provides a principled explanation for related research dedicated to estimate the placement of the applicator without MRI. To further demonstrate the framework's potential for clinical applications, we implemented it into a markerless robotic neuronavigation system and validated its performance on both phantoms and insomniac patients. By bypassing optical markers, this system overcomes issues such as marker displacement and operator dependence[10,14-18] through markerless positioning to simplify clinical procedures and automate the workflow[10, 34, 35].

By integrating fragmented anatomical correlations into the unified framework, this paradigm overcomes the limitations of MRI while ensuring the accuracy, reproducibility and explainability of the method. Additionally, it reduces healthcare costs and the time required for patients while minimizing manual operation, thereby facilitating the broader application of non-invasive neuromodulation as a non-pharmacological treatment across multilevel healthcare settings. For instance, in large hospitals or neurology-specialized institutions, this framework would partially alleviate issues like prolonged patient wait times and physician workloads stemming from strained medical resources[10]. In psychiatric clinics, where independent MRI scanning is not feasible, neuromodulation treatment often relies on manual localization, making it difficult to ensure targeting accuracy and reproducibility, and thus limiting therapeutic efficacy[12,13]. The introduction of this framework can provide patients with more reliable treatment. Neuromodulation has emerged as an increasingly preferred approach for select psychiatric and neurological disorders in community hospitals or rehabilitation centers[34-36]. Compared to traditional neuronavigation, this framework presents a lower barrier to use, thereby enables more patients to conveniently access non-invasive neuromodulation therapy. If this framework can be widely adopted in community and primary healthcare settings, it may even support the future development of automated, unmanned neuromodulation services.

To translate this decentralized clinical concept into practical application, we present and validate an automated workflow for non-invasive neuromodulation via a markerless robotic neuronavigation system. The workflow utilizes a face-scalp-cortex mapping framework to guide target localization, with the robotic system executing the placement of the applicator. Targeting accuracy was evaluated in silico and on phantom. The workflow was then used to administer TMS to patients with insomnia and show favorable therapeutic outcomes. This MRI-free, markerless non-invasive neuromodulation paradigm enables the expansion of treatment to multi-level healthcare institutions.

Results

Automated workflow for non-invasive neuromodulation

The initial treatment workflow of the robotic navigated neuromodulation involves a series of automated processes (Fig. 1a):

step1, face-to-scalp mapping: Scan facial point clouds, remove non-facial points and hair, and optimize raw point clouds to obtain smooth facial point clouds. Upload the point clouds to the cloud server, where the cranial surface generative modeling completes the full head point clouds. The model is then transmitted back to the local neuronavigation software, which reconstructs the 3D head model.

step2, cortex-to-scalp mapping: Determine the MNI coordinates for the treatment target. Mark the model's nasal root point and the anterior points of both ears. Calculate the applicator placement positions on the scalp using cortex-to-scalp mapping.

step3, markerless robotic neuronavigation: For the initial treatment, mark the midpoints of the left and right eyeballs and the nasal tip on the model. Under the guidance of camera, align the model with the actual head via system registration. The robot positions the applicator on the participant's head according to the planned target and maintains head-applicator relative pose via contact force throughout the session, thereby ensuring safety and secure adhesion.

In this study, 18 patients with insomnia received TMS treatment using the aforementioned workflow (Fig. 1b). Due to facial privacy protection measures implemented in Step 2, for first-time patients, the entire process from facial scanning to treatment device activation requires approximately 10-15 minutes. At present, facial scanning, point cloud segmentation, and model marker selection are carried out by the operator. While the robot is capable of tracking head motion during treatment and performing safe, force-controlled motion compensation, it will retract the applicator and suspend stimulation in the event of sudden or excessive head movements.

The robotic navigation system consists of a hardware platform and a software suite (Fig. 1c), including a handheld 3D scanner to acquire facial point clouds. The system is compatible with several non-invasive neural modulation device, such as TMS, TUS, or tPBM. A 6-DOF robotic arm carries the applicator with a 6-axis force/torque sensor at the flange end. A stereo camera captures facial landmarks and calculates their 3D coordinates. The navigation software performs 3D head model reconstruction and rendering, target coordinate selection and mapping, spatial registration, robotic execution planning, and system status monitoring.

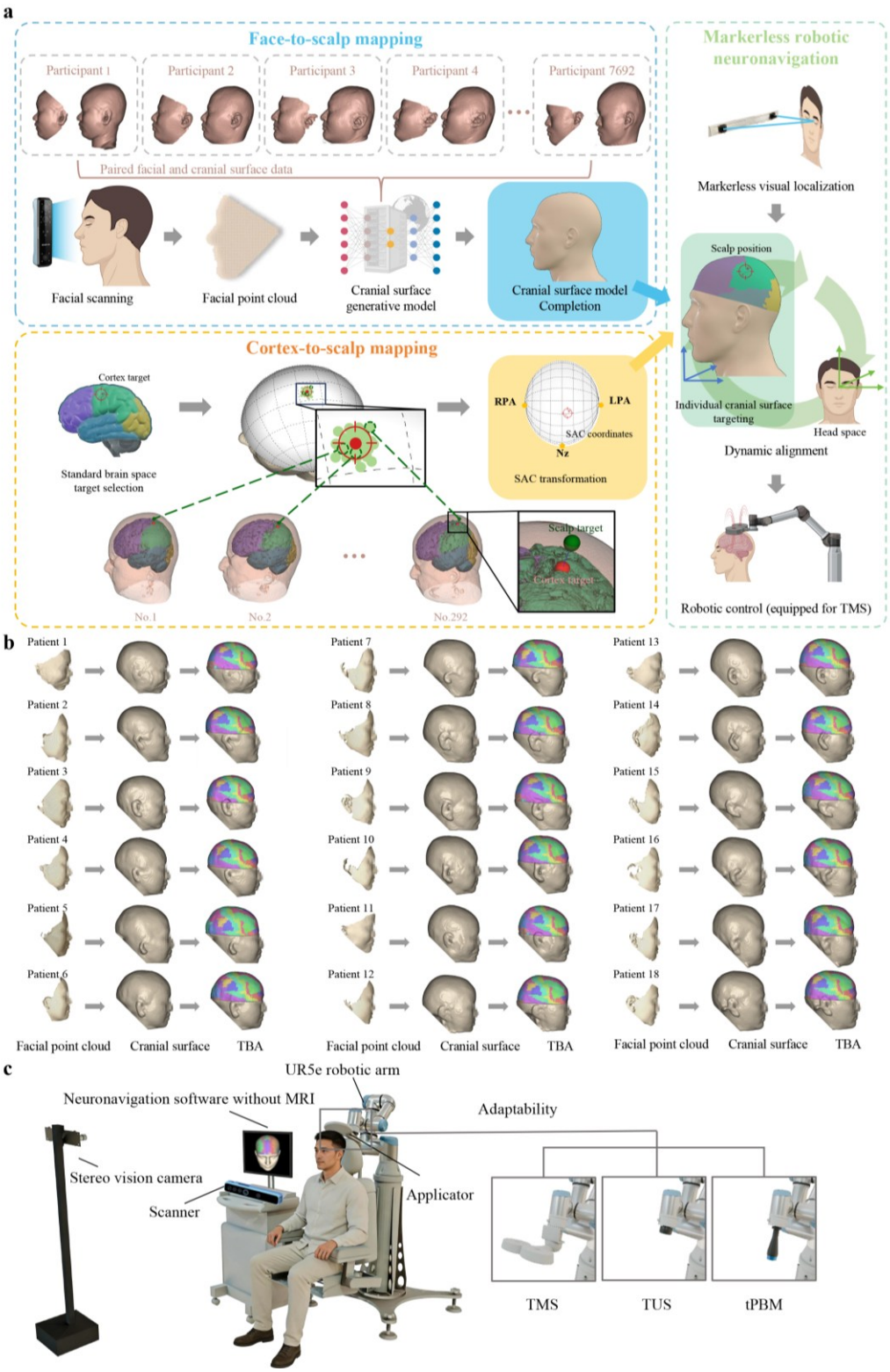


Fig. 1 | Automated workflows and system composition. a. The workflow for the first-time neuromodulation treatment. During the face-to-scalp mapping phase, facial point cloud is acquired via scanner. The cranial surface generative modeling was trained using 7,692 pairs of facial and cranial surface data. To protect privacy, the cranial surface is reconstructed on cloud servers and transmitted back to the local neuronavigation system. During the cortex-to-scalp mapping phase, selected brain target is mapped onto the generated cranial surface model using the scalp angle coordinate system (SAC). The mapping is determined by 292 individual scalp positions. In the markerless robotic neuronavigation phase, stereo vision camera guides dynamic registration of the neuronavigation system, culminating in robotic arm-assisted applicator placement. **b.** The facial point clouds collected from 18 insomnia patients were processed using a hierarchical face-scalp-cortex mapping and the mapping results are displayed on the transcranial brain atlases (TBA). **c.** Overview of markerless robotic navigated non-invasive neuromodulation system. The neuromodulation system operates without MRI guidance and comprises hardware modules including a scanner, stereo vision camera, and robotic arm. Non-invasive neuromodulation devices can be adapted to utilize transcranial magnetic stimulation (TMS), transcranial ultrasound (TUS), or transcranial photobiomodulation (tPBM).

Face-to-scalp mapping based cranial surface reconstruction

We discovered the latent correlation between facial and cranial surfaces (Fig. S1), thereby developing a face-to-scalp mapping. By utilizing facial information as prompt input through a generative modeling, we successfully reconstructed complete cranial surfaces. In the generative model, the variational autoencoder (VAE) is used for topological representation, while the diffusion model is used to map facial topological representations to cranial topological representations. We constructed a dataset of 8,195 participants with complete facial and cranial coverage on three-dimensional T1-weighted MRI (3D-T1WI). Of these, 503 cases were reserved as the test set, and the remaining participants were used to train the generative modeling. For each test case, we performed 3D surface reconstruction on the 3D-T1WI, segmented the facial region, and fed the segmented data into the generative model to obtain the reconstructed cranial surface (Fig. 2a). Using MRI-reconstructed cranial surfaces as the gold standard, we evaluated the generated surfaces using metrics including mean distance, chamfer distance (CD), earth mover's distance (EMD), and normal consistency (NC) (Table 1). To evaluate the completion accuracy across different scalp regions, cranial surfaces were mapped onto a scalp angle coordinate (SAC) system constructed using the nasion (Nz), right preauricular point (RPA), and left preauricular point (LPA) (Fig. 3a). For most regions, the average reconstruction error of the completion model was less than 1.05 mm, with larger errors observed in the occipital area (Fig. 2b, f). The NC exhibited a more uniform error distribution; however, errors were higher in regions near the RPA and LPA due to the anatomical complexity of the ears. These areas, nonetheless, were beyond the scope of the present study (Fig. 2c, g).

A dataset was constructed by acquiring facial point clouds with a 3D scanner and pairing them with corresponding 3D-T1WI data (n=102). The cranial surface generative modeling, initially trained on the MRI dataset, was then fine-tuned by the dataset to provide complete cranial surface model for the workflow without MRI (Fig. 2a). To evaluate the clinical performance of the generative modeling, 18 patients with insomnia were recruited. Larger mean surface distances were observed at scalp regions near the nasion (Nz) on the forehead and in the occipital area (Fig. 2d). Compared with MRI-based generated cranial surface model, most scalp regions showed larger mean distances and higher NC errors; however, certain frontal and parietal regions still exhibited considerable accuracy (Fig. 2e–g) (n = 18).

Additionally, to evaluate the impact of variations in facial segmentation among different operators during practical use, we had three operators independently perform facial segmentation on 503 cases from the MRI test set. We then calculated metrics for the three groups of generated cranial surface models (Table 1). To demonstrate the consistency of the results, we computed the intra-class correlation (ICC) for four metrics: mean distance, CD, EMD, and NC. The ICC (2,3) for mean distance was 0.89 (95% CI: 0.81–0.93); the ICC (2,3) for CD was 0.91 (95% CI: 0.88–0.94); The ICC (2,3) for EMD was 0.65 (95% CI: 0.59–0.70); the ICC (2,3) for NC was 0.71 (95% CI: 0.41–0.84). When different operators perform facial segmentation, the cranial surface generative modeling exhibits excellent absolute consistency.

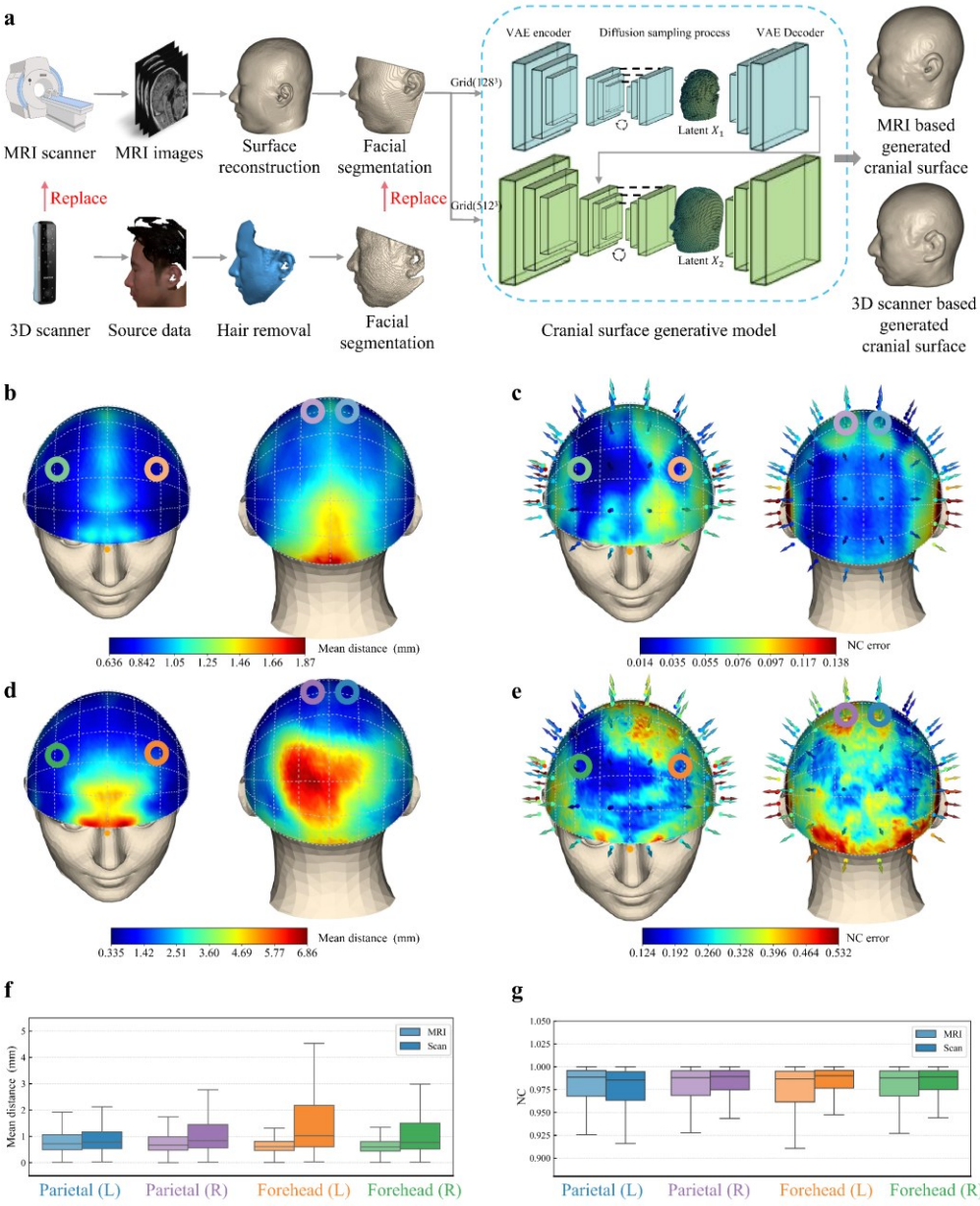


Fig. 2 | Face-to-scalp mapping based cranial surface reconstruction. a. A cranial surface generative modeling conditioned on facial point cloud employs a coarse-to-fine hierarchical architecture ($128^3 \rightarrow 512^3$ resolutions). Leveraging extensive MRI datasets, the pre-trained model learns latent relationships between facial and cranial structures. In practical applications, facial point clouds are acquired using scanners instead of MRI devices. **b.** Spatial distribution of the mean distance of point clouds calculated between the MRI-based generated cranial surface model and the MRI reconstruction model. (n=503) **c.** Spatial distribution of the normal consistency (NC) error of point clouds calculated between the MRI-based generated cranial surface model and the MRI reconstruction model. The NC error is calculated as 1 minus the product of the corresponding unit normal vectors from the two models. (n=503) **d, e.** Spatial distribution of the mean distance and NC error of point clouds calculated between the scanner-based generated cranial surface model and the MRI reconstruction model. (n=18) **f.** The mean distance distribution of the cranial surface regions circled in (b) and (d). **g.** The NC distribution of the cranial surface regions circled in (c) and (e). The top, middle, and bottom boundaries of the boxplots represent 25th, 50th, and 75th percentile, respectively; Outliers defined when value larger than 1.5*IQR + 75th percentile.

Table 1 Evaluation metrics for the generated cranial surfaces

	Mean distance	CD	EMD	NC
MRI (operator 1)	1.05±0.37	2.42± 0.61	1.63± 0.02	0.93 ± 0.01
MRI (operator 2)	1.21±0.53	2.59± 0.74	1.63± 0.02	0.92 ± 0.01
MRI (operator 3)	1.32±0.51	2.69± 0.72	1.63± 0.02	0.92 ± 0.01
Scan	2.56±0.81	5.84±1.64	1.62±0.01	0.92±0.017

Cortex-to-scalp mapping for simulation targeting *in silico*

A cortex-to-scalp mapping is constructed to locate brain targets (MNI coordinates) and transform them onto the scalp. Unlike existing CPC methods[20,24] based on relative lengths of the scalp geodesic, we propose a SAC using angle coordinate (Fig. 3a, Fig. S2a) to accommodate scenarios where facial point clouds only capture three points: Nz, RPA, and LPA. The mapping was constructed using the publicly available SLIM dataset of healthy individuals[37]. To evaluate the performance of the SAC-based cortex-to-scalp mapping, we compute the scalp targeting error and cortical targeting error for a given MNI cortical target (Fig. 3b). For the healthy participants in the SLIM dataset (n = 292), a leave-one-out approach was employed: for each of the 292 participants, the cortex-to-scalp mapping was constructed using the remaining 291 participants when calculating the error. The calculated mean scalp targeting errors ranged from 2.36 mm to 5.70 mm (Fig. 3c, d), while mean cortical targeting errors ranged from 2.50 mm to 8.81 mm (Fig. 3e, f). Since the cortex-to-scalp mapping was constructed based on healthy participant data, it is necessary to validate its errors in patients and further validate the errors in the hierarchical face-scalp-cortex mapping. Mapping errors were calculated on the MRI-reconstructed surfaces and generated cranial surfaces of 18 insomnia patients. The average scalp targeting error ranged from 3.08 mm to 8.19 mm and from 4.80 mm to 14.00 mm (Fig. 3c, d), while the average cortical targeting error ranged from 2.39 mm to 9.84 mm and from 2.05 mm to 8.12 mm (Fig. 3e, f). Across the three datasets, errors generally increased from anterior to posterior regions. The generated cranial surface exhibited the highest mean scalp targeting error, but showed no significant difference in mean cortical targeting error compared to the other two datasets (Fig. 3c, e). Specifically, comparing the distribution of cortical targeting errors across the three datasets in the primary motor cortex (M1) and dorsolateral prefrontal cortex (DLPFC)—common non-invasive neuromodulation treatment regions—revealed no significant differences (Fig. 3f).

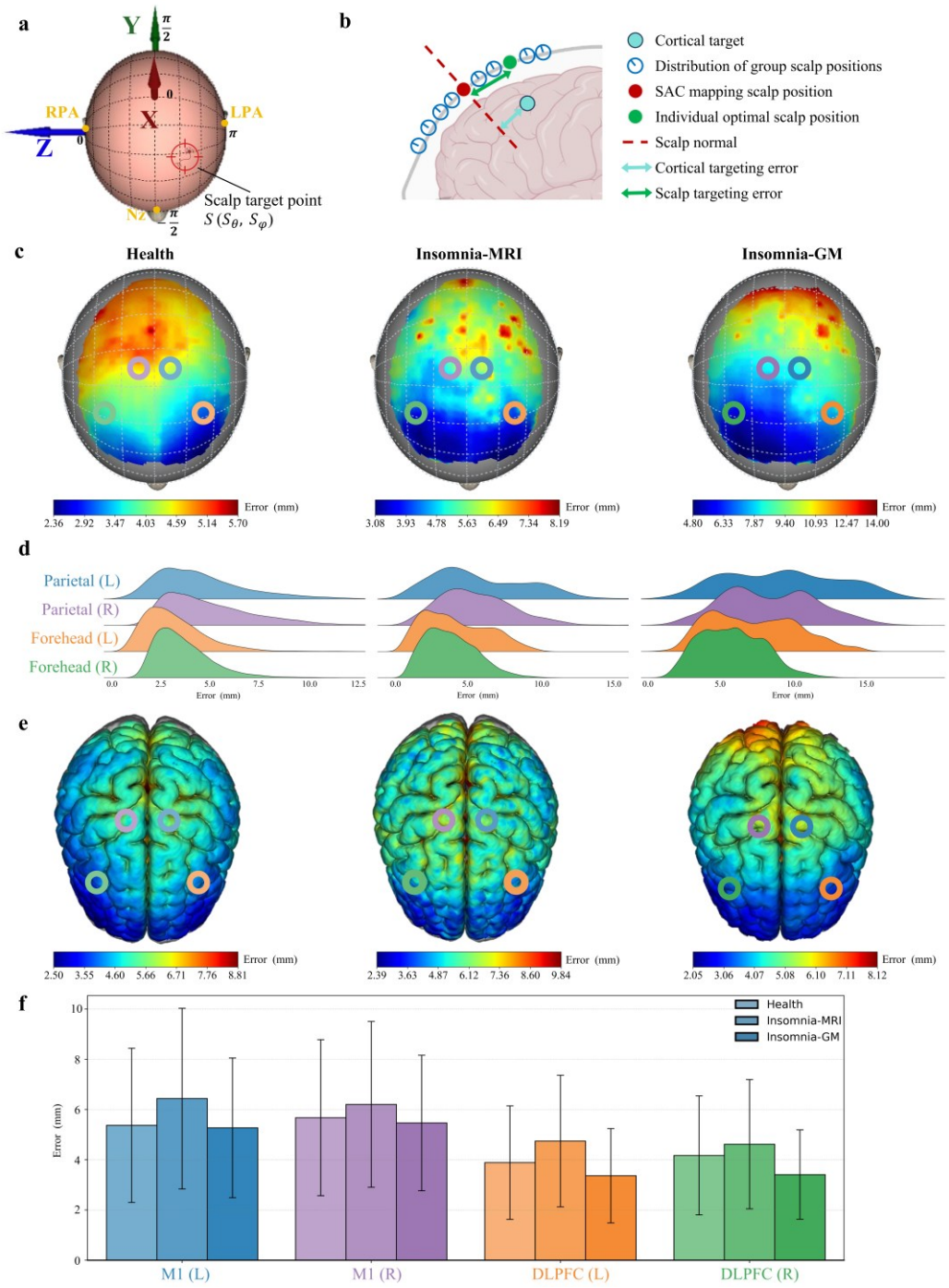


Fig. 3 | Cortex-to-scalp mapping. a. Definition of SAC. Any scalp point can be represented as $S(S_\theta, S_\varphi)$, where $S_\theta \in [0, \pi], S_\varphi \in [-\pi/2, \pi/2]$. **b.** Definition of scalp targeting error and cortical targeting error. The SAC mapping scalp position is obtained from the statistical distribution of group scalp positions. Scalp targeting error is defined as the distance from the SAC mapping scalp position to the test participant's individual optimal scalp position. Cortical targeting error is defined as the perpendicular distance from the test participant's cortical target position to the scalp normal at the SAC mapping scalp position. **c, e.** Spatial distribution of scalp targeting error and cortical targeting error on MRI reconstruction models of healthy participants (n=292), insomnia patients (n=18), and generated cranial surface models of insomnia patients (n=18). **d.** Distribution of scalp targeting errors in the scalp region circled in (c). **f.** Distribution of cortical targeting errors in the cortical region circled in (e). Error bars represent mean ± SD.

Validation of markerless robotic neuronavigation on phantom

To enhance the accessibility of the hierarchical face-scalp-cortex mapping, we have incorporated markerless robotic neuronavigation system based on facial landmarks into the workflow. Unlike

marker-based navigation methods, AI-derived facial landmarks exhibit inherent uncertainty fluctuations. By dynamically adjusting the 3D model landmarks, more precise registration between camera space S_{cam} and model space S_{render} is achieved (Fig. 4a, b). To verify the positioning accuracy, a 3D-printed head model featuring target holes was produced (Fig. S3). Under MRI guidance, a robotic arm performs applicator placement. Target point position errors and orientation errors are compared between the unadjusted landmarks method and landmarks position dynamic optimization method. The optimization method improved the accuracy of target positioning, demonstrating better stability and repeatability; the mean positional error for this method was less than 5 mm, and the mean angular error was less than 3° across all sampling points (Fig. 4c). Next, the positioning accuracy of this method is validated by the navigation system without MRI, in which the results of the hierarchical face-scalp-cortex mapping were used as target reference coordinates. The DLPFC exhibits smaller positional and angular errors compared to M1 and PM. The positioning errors at most sampling points are less than 10 mm (Fig. 4d). For safety reasons, compliant control is employed during treatment, maintaining applicator-to-head contact force near 4 N (Fig. 4e) while ensuring tight applicator adhesion to the scalp.

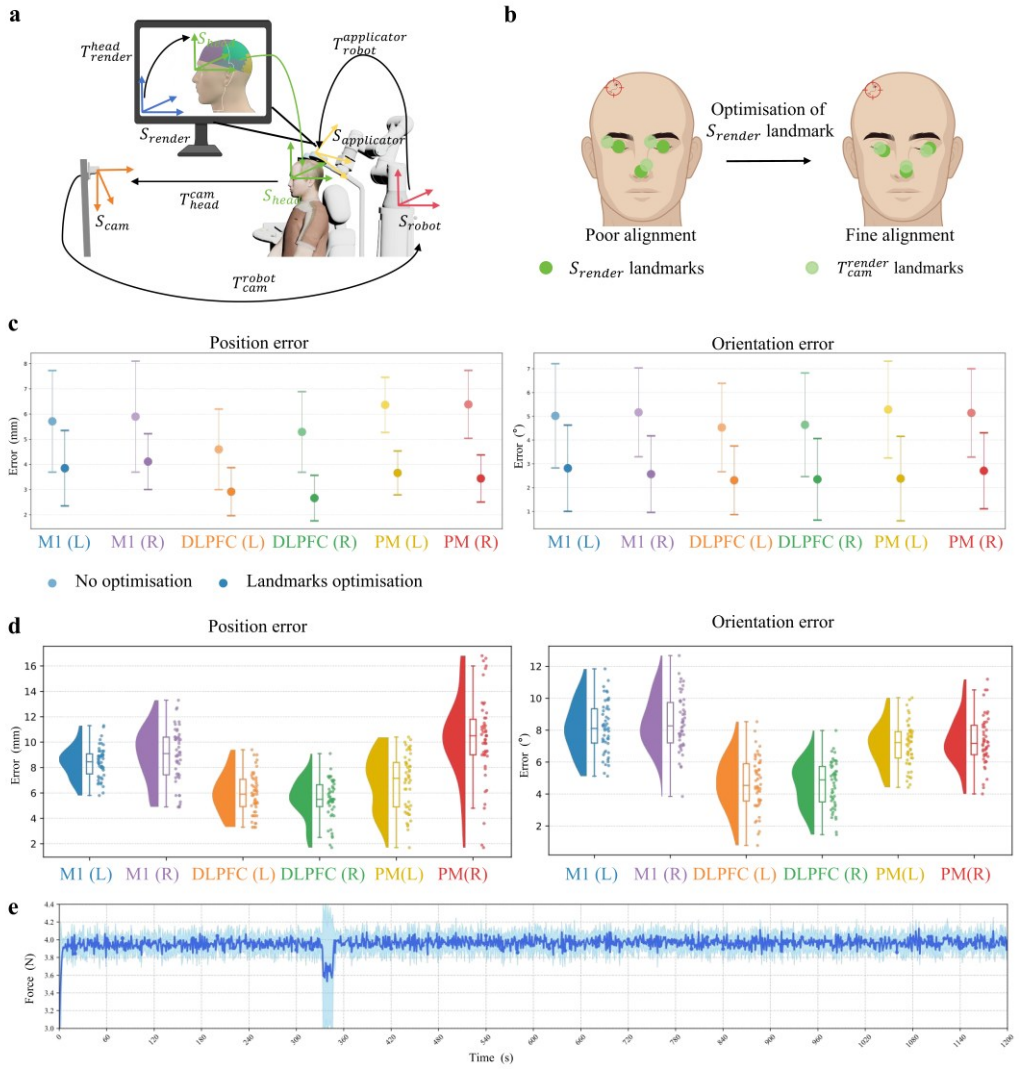


Fig. 4 | Validation of markerless robotic neuronavigation on phantom. a. System coordinate registration. The virtual head model and the real participant's head share the same coordinate system S_{head}. **b.** Optimization of virtual head model alignments with real participant' head. By refining the landmarks of the virtual head model based on facial landmarks detected by a stereo vision camera, achieving improved alignment results. **c.** MRI-guided positioning and orientation errors. Error bars represent mean ± SD. **d.** MRI-free positioning and orientation errors. The top, middle, and bottom boundaries of the boxplots represent 25th, 50th, and 75th percentile, respectively; Outliers defined when value larger than 1.5*IQR + 75th percentile. **e.** Contact force (z axis) between the participant's

head and the applicator (n=10). The shaded area represents mean ± SD.

Validation for insomnia *in vivo*

A clinical validation study was conducted using the automated workflow without MRI guidance. 18 patients with insomnia received ten sessions of 1 Hz rTMS treatment over a two-week period (Fig. 5a). Sleep quality was measured before and after treatment using the Pittsburgh Sleep Quality Index (PSQI) and Insomnia Severity Index (ISI). Results indicate a significant improvement in sleep quality after treatment. The mean PSQI score decreased from 13.39 (before treatment) to 8.39 (after treatment) with 10 participants showing a reduction of at least 5 points (Fig. 5b) (n=18). The mean ISI score decreased from 18.56 to 12.17 with 11 patients showing a reduction of at least 5 points (Fig. 5f) (n=18).

Moreover, to validate the impact of the estimation of target locations using the workflow on the efficacy of TMS treatment for insomnia, we categorized patients into Group 1 (improvement < 5 points) and Group 2 (improvement ⩾ 5 points) based on pre-to-post scale changes. We further assessed the distribution patterns of three mapping index—mean distance, scalp targeting error, and cortical targeting error—between the two groups (Fig. 5c-e, g-i). Notably, the estimation errors for targeting in the two groups showed no significant correlation with differences in therapeutic efficacy.

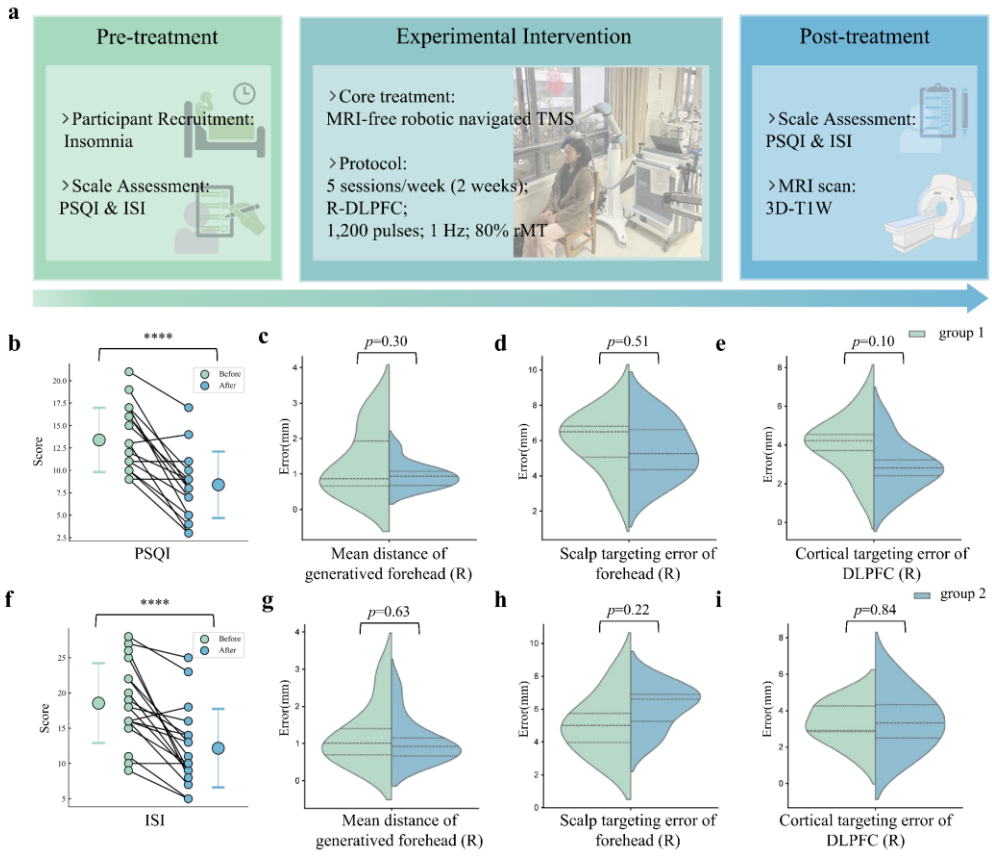


Fig. 5 | Validation for insomnia in vivo. a. Experimental Procedure. 18 insomnia patients underwent two weeks of robotic navigated TMS treatment without MRI guidance, targeting the right dorsolateral prefrontal cortex (R-DLPFC; MNI: x=44, y=38, z=34). **b.** Changes in PSQI scores before and after treatment (n=18). p-values obtained via paired t-tests. ****p<0.0001. **c-e.** Assign participants with PSQI scores reduced by less than 5 points to group 1, and the remainder to group 2. Compare whether there are between-group differences in the mean distance of generated right forehead, scalp targeting error of right forehead, and cortical targeting error of R-DLPFC. The p-value is obtained using Welch's t-test. **f.** Changes in ISI scores before and after treatment (n=18). p-values obtained via paired t-tests. ****p<0.0001. **g-i.** Assign participants with ISI scores reduced by less than 5 points to group 1, and the remainder to group 2. Compare whether there are between-group differences in the mean distance of generated right forehead, scalp targeting error of right forehead, and cortical targeting error of R-DLPFC. The p-value is obtained using Welch's t-test.

Discussion

According to statistics, approximately 15% of the global population has a mental disorder[38], and more than 3 billion people are affected by a neurological condition[39], imposing significant burdens on society and families while impacting individual life quality. Non-invasive neuromodulation technique, has gained increasing application in treating these conditions in recent years[1-8]. This therapeutic approach can help reduce patients' dependence on medication to some extent. Neuronavigation has provided reliable methods for precise treatment targeting[10,16,17]. However, their reliance on MRI-based guidance has limited their applicability across a broader range of disorders and clinical settings[12,13,20,22,24]. While several studies have explored MRI-free methods[20,22-27], they view geometric surface reconstruction and surface-to-brain mapping as isolated research domains. Here, we propose a unified framework workflow for non-invasive neuromodulation via a markerless robotic neuronavigation system (Fig. 1) that bridges this gap through a hierarchical face–scalp–cortex mapping. The system utilizes a face-to-scalp mapping (Fig. 2) and a cortex-to-scalp mapping (Fig. 3) to replace patient MRI image data, thereby reducing medical costs for patients. Additionally, the navigation system employs automatic and markerless localization (Fig. 4) to decrease reliance on specialized physician operation. This workflow demonstrates greater suitability and alignment with available medical resources in settings such as psychological clinics and community hospitals, possessing significant potential for widespread adoption.

Several existing surface-to-brain mapping studies utilize facial landmarks to predict either the neurocranial landmarks of the International 10–20 System or specific brain targets[25-27]. Although these approaches have demonstrated promising predictive accuracy, they frequently lack a principled explanation, leaving a theoretical loophole regarding the modeling rationale of direct face-to-brain mapping. To bridge this gap, we introduce a geometric surface reconstruction method and develop a hierarchical face–scalp–cortex mapping framework. Specifically, the VAE is employed to extract the topological latent representations of both facial and cranial surfaces, revealing the correlation between these latent spaces (Fig. S1). The diffusion model is leveraged to learn the mapping from the facial latent representation to its cranial latent representation. We also evaluated the mean distance and NC between cranial surface models constructed using face-to-scalp mapping and MRI-based models, as these metrics influence the targeting accuracy. The proposed cortex-to-scalp mapping method employed three anatomical landmarks—Nz, LPA, and RPA—and adopted a SAC system analogous to geographic latitude and longitude (Fig. 3a). This design not only accommodates the scanner's ability to accurately capture only facial point clouds, but also decouples the mapping construction from scalp spatial information. Compared to previous methods that employed scalp geodesics for mapping modeling[20,24], this approach effectively prevents the further accumulation of errors between mapping construction and cranial surface model reconstruction (Fig. S2a). Currently, the proposed surface-to-brain mapping framework still faces several limitations. First, the paired dataset of facial scans and MRI images used for fine-tuning the cranial surface generation model remains limited. Future work will focus on collecting more paired data for training and incorporating a domain adaptation module to mitigate the impact of input discrepancies caused by deformations between the MRI facial point cloud and the scanner facial point cloud on the face-to-scalp mapping. Second, in constructing the cortex-to-scalp mapping, this study followed a common assumption that the scalp point with a normal vector most closely aligned with the cortical target represents the optimal applicator placement for

neuromodulation (the linear model [40]). However, considering the spatial distribution characteristics of physical fields within the complex cortical structures[41-44], this may not always correspond to the true optimal targeting pose of the applicator. In future, stimulation simulations tailored to specific neuromodulation device could be employed to establish a more precise mapping.

Unlike existing commercial robotic neuronavigation systems[10,14-18], we employ a markerless approach for head positioning. This approach aims to minimize manual intervention and achieve a higher degree of automation, targeting primary community hospitals or unmanned medical settings. So the automated workflow for non-invasive neuromodulation via the markerless robotic navigation system will enable patients to access professional neuromodulation during the early stages of illness, facilitating timely relief or cure for mild symptoms while reducing medication usage and abuse. The robotic navigation system enhances the clinical value of the face-scalp-cortex mapping framework. The proposed navigation system focuses on ensuring consistency between the 3D model space and camera space registrations. Compared with manual annotation, AI-based recognition of facial landmarks exhibits greater uncertainty and instability, which amplifies registration errors between the landmarks detected by the camera and those on the head model. Our approach dynamically adjusts landmarks within the head model space, significantly reducing transformation errors between S_{cam} and S_{render} spaces. The variance was reduced by 2.1 mm compared with the non-optimized method, demonstrating the stability and reproducibility of the proposed approach for markerless localization of guided but hard-to-identify target points (Fig. 4c). Stereo camera exhibits inherent limitations in 3D spatial accuracy and is susceptible to environmental lighting conditions. Future approaches may employ 3D point cloud cameras equipped with near-infrared/structured light sensors for markerless localization[45,46]. High-precision point cloud cameras acquire facial point clouds with minimal error and are less affected by ambient light. Compared with existing methods, they offer greater robustness by enabling high-density point cloud acquisition for localization and spatial registration. However, addressing challenges related to point cloud matching consistency and real-time performance will remain a key issue.

Using the proposed workflow, a two-week treatment was conducted on 18 sleep disorder patients. Significant improvements in PSQI and ISI scores were observed in most patients post-treatment, demonstrating favorable therapeutic outcomes and clinical utility of the workflow. Furthermore, comparative analysis across patient groups confirmed that estimation errors had no significant impact on treatment efficacy. A limitation of this study is the relatively small in vivo sample size. Future work will include a broader range of diseases and patients. In addition, to enhance the accessibility of the proposed system, future efforts will focus on miniaturizing or making robotic devices wearable[15], thereby further enabling their application in community and home settings.

In summary, this study overcomes the limitation regarding manual operation and model explainability in current MRI-free methods. The results indicate that using a face-scalp-cortex mapping framework to estimate the neuromodulation applicator placement pose yields targeting errors within the acceptable range of the effective stimulation spread of neuromodulation in silico. This demonstrates the feasibility of planning targeting therapeutic interventions without MRI guidance. Additionally, the markerless robotic navigation system increases the accessibility of the targeted positioning guided by the framework. In vivo application of the automated workflow in insomnia patients shown favorable

therapeutic outcomes. Given its aforementioned advantages, the workflow shows strong potential for implementation in community healthcare settings, facilitating timely and accessible non-invasive neuromodulation treatment and supporting early non-pharmacological intervention for patients with mild conditions.

Materials and Methods

Data acquisition and human participants

A 3.0T MRI database comprising 8,195 adult participants (aged 18–70 years, 4166 men, 4029 women) is utilized for training and validating cranial surface generative modeling. This database includes 3D-T1WI scans collected over a 10-year period (2015–2024) at the First Affiliated Hospital of Sun Yat-sen University. A total of 102 paired facial point clouds and MRI data are collected from the department of radiology at the hospital. Facial point cloud data are acquired using an Einstar scanner, classified as Class I laser safety (eye-safe), with point spacing sets to 0.5 mm. The collection of data is conducted under the supervision of the Ethics Committee of the First Affiliated Hospital of Sun Yat-sen University (IIT-2024-949), to ensure the data anonymization. Collected MRIs are required to have intact skulls and faces; images are excluded if patients show a history of brain trauma, prior craniotomy, or imaging artifacts due to dental prostheses, facial swelling, or other abnormalities. The MRI data are acquired with a resolution of ≤1 mm in all directions. During post-processing, the images are resampled to achieve isotropic voxels of $1 \times 1 \times 1$ mm³, followed by surface reconstruction and extraction. Facial scan point clouds are also converted to voxel data with a resolution of $1 \times 1 \times 1$ mm³. Both MRIs and facial scan point clouds are then voxelized into coarse and fine grids at resolutions of 128^3 and 512^3, respectively, for model training.

We select the 3D-T1WI data from 292 young Chinese adult participants in the SLIM dataset[37] to construct cortical-to-scalp mapping. The images are obtained using a magnetization-prepared, rapid acquisition gradient-echo sequence (echo time (TE) = 2.52ms, repetition time (TR) = 1900ms, flip angle = 9°, 1×1×1 mm isotropic voxels).

To further validate the positioning errors and clinical practicality of the workflow in non-invasive neuromodulation therapy, we recruit 18 patients (aged 20–63 years, 4 men, 14 women) with insomnia for in-vivo experiment. This study is conducted under the approval of the Ethics Committee of the Second Affiliated Hospital of South China University of Technology (K-2022-069-04) and with informed consent from each participant. Recruitment bias primarily stems from limitations in the model training data sources and geographic scope, restricting participants to specific regions within China.

Generative modeling for face-to-scalp mapping

To construct the face-to-scalp mapping, we employ a large-scale 3D generative modeling for cranial surface completion (Fig. 2a). Based on a high-resolution unconditional hierarchical generative modeling architecture[30], we extend this approach to the task of cranial surface completion with facial point clouds as conditional input. The cranial surface generative modeling comprises two levels, coarse-to-fine, each containing two modules: the sparse structure VAE and the latent diffusion model (LDM). The sparse structure VAE learns shape priors for the head and face to construct latent

representations. At the coarse level, the LDM learns the latent representation of the head, conditioned on the facial latent representation. At the fine level, the LDM learns the high-resolution latent representation of the head, conditioned on the coarse level.

During the training of the level-l sparse structure VAE, we employ the following loss function:

$$\mathcal{L}_l^{\mathrm{VAE}} = \mathbb{E}_{\mathbf{H}_l}\left[\mathbb{E}_{\mathbf{X}_l\sim q_\phi}[\lambda_1 \mathrm{BCE}(\mathbf{H}_l, \widetilde{\mathbf{H}}_l) + \lambda_2 ||n - n_{GT}||_2^2] + \lambda_3 \mathbb{KL}(q_\phi(\mathbf{X}_l)||p(\mathbf{X}_l))\right]$$

Where $\mathbf{H}_l$ is the voxel grid of the cranial surface, and $\widetilde{\mathbf{H}}_l$ is the output of the VAE decoder given $\mathbf{X}_l$. $\mathrm{BCE}(\mathbf{H}_l, \widetilde{\mathbf{H}}_l)$ denotes the binary cross-entropy loss measuring the discrepancy between the reconstructed shape $\widetilde{\mathbf{H}}_l$ and the ground-truth shape $\mathbf{H}_l$. $||n - n_{GT}||_2^2$ computes the mean squared error between the reconstructed cranial surface normals n and the ground-truth normals n_{GT} to ensure geometric detail accuracy. $\mathbb{KL}\left(q_\phi(\mathbf{X}_l)||p(\mathbf{X}_l)\right)$ represents the Kullback–Leibler divergence loss that constrains the latent distribution $q_\phi(\mathbf{X}_l)$ to approximate the prior distribution $p(\mathbf{X}_l)$, thereby enhancing the regularization of the latent space. λ_1, λ_2, and λ_3 are the weighting factors for the three loss terms. The face constitutes a subset of the cranial surface. Upon voxelization, the latent representations output by the VAE encoder correspond to this local structure, eliminating the need to train another VAE specifically for facial data.

At the coarse level of the LDM, an interleaved attention mechanism is introduced to guide the facial latent vector $\mathbf{F}_0$ toward the head latent vector representation $\mathbf{X}_0$. At each denoising step t, given the current noisy head latent representation $\mathbf{X}_0^t$, we treat it as the query Q_t, while $\mathbf{F}_0$ is linearly transformed into the key K_t and value V_t. The interleaved attention mechanism computes the similarity between Q_t and K_t and applies a weighted sum over V_t:

$$\mathrm{Attention}(Q_t, K_t, V_t) = \mathrm{softmax}\left(\frac{Q_t K_t^T}{\sqrt{d_k}}\right)V_t$$

Then the output of the interleaved attention $\mathbf{X}_0^{t,\mathrm{attn}}$ is obtained by fusing the original latent representation with the attention-weighted results, providing the updated latent representation for the next denoising step:

$$\mathbf{X}_0^{t,\mathrm{attn}} = \mathbf{X}_0^t + \mathrm{Attention}(Q_t, K_t, V_t)$$

The loss function for training the LDM is:

$$\mathcal{L}_l^{\mathrm{LDM}} = \mathbb{E}_{\mathbf{C}_l, t, \mathbf{X}_l, \boldsymbol{\varepsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I})}\left[\| \boldsymbol{v}_{\boldsymbol{\theta}_l}(\mathbf{X}_{l,t}, t, \mathbf{C}_l) - \boldsymbol{v}_{\mathrm{ref}} \|_2^2\right]$$

where $\boldsymbol{v}_{\boldsymbol{\theta}_l}$ is the denoising network conditioned on $\mathbf{C}_l$, $\boldsymbol{v}_{\mathrm{ref}} = \sqrt{\bar{\alpha}_t}\boldsymbol{\varepsilon} - \sqrt{1-\bar{\alpha}_t}\mathbf{X}_l$, $t \sim [1, T]$, $\mathbf{X}_{l,t} = \sqrt{\bar{\alpha}_t}\mathbf{X}_l + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\varepsilon}$, and $\bar{\alpha}_t$ is a time-step-dependent variance scheduling function used to control noise injection. $\mathbf{C}_l$ in the coarse and fine levels corresponds respectively as follows:

$$\mathbf{C}_l = \begin{cases} \mathbf{F}_0, & l = 0 \\ \widetilde{\mathbf{H}}_{l-1}, & l = 1 \end{cases} .$$

In silico studies on cortex-to-scalp mapping

The 3D-T1WI data are segmented and the cranial surface reconstructed using the charm pipeline in SimNIBS[47]. The optimization tool within SimNIBS is employed to obtain optimized scalp coordinates

for designated cortical targets. The optimization aims to minimize the distance between the cortical target and the line of the scalp point normal. Scalp point searches are confined to a 1cm radius around the scalp point closest to the cortical target, encompassing approximately 600 scalp points.

The scalp angle coordinate (SAC), a spherical coordinate, is constructed using three points: the nasion (Nz), and the left/right preauricular points (LPA/RPA) (Fig. 3a). The Cartesian origin O is set at the projection of the nasion (Nz) onto the LPA–RPA line (i.e., the foot of the perpendicular from Nz to the line). The Z-axis lies along the LPA–RPA line, the Y-axis along the O–Nz line, and the X-axis is perpendicular to the YOZ plane. Based on this, the SAC is defined, allowing any scalp target point to be represented as $S(S_\theta, S_\varphi)$, where S_θ is the angle between the positive Z-axis direction and $\overrightarrow{OS}$ ($S_\theta \in [0, \pi]$), S_φ is the angle between $\overrightarrow{OS'}$ and the X-axis ($S_\varphi \in [-\frac{\pi}{2}, \frac{\pi}{2}]$), and S' is the projection of S onto the XOY plane.

For a specified MNI coordinate on the cortex, optimized scalp coordinates are obtained for each participant and transformed to the SAC, denoted as $S^k(S_\theta^k, S_\varphi^k)$. The cortex-to-scalp mapping corresponding to these MNI coordinates is then calculated via phase averaging (Fig. S2b), with the mathematical expression given as follows:

$$S = (arg\left(\frac{1}{N}\sum_{k=1}^{N} e^{iS_\theta^k}\right), arg\left(\frac{1}{N}\sum_{k=1}^{N} e^{iS_\varphi^k}\right))$$

where N is the number of participants, $arg(\cdot)$ is the argument of a complex number, i is the imaginary unit.

Coordinates registration and visual orientation

To achieve automated positioning of the applicator, it is necessary to establish coordinate registration between the rendering coordinate system S_{render}, head coordinate system S_{head}, camera coordinate system S_{cam}, robot base coordinate system S_{robot}, and applicator coordinate system $S_{applicator}$ (Fig. 4a). The position of the applicator relative to the robot base $T_{robot}^{applicator}$ is calculated using the forward kinematics, with the tool center point and robot flange offset. The transformation T_{cam}^{robot} between S_{robot} and S_{cam} is obtained through hand-eye calibration. Spatial coordinates of the participant's facial landmarks are captured in real-time via a stereo vision camera, establishing the transformation T_{head}^{cam} between the camera and the head. T_{render}^{head} is used to align the space in the rendering software, the visualization toolkit (VTK), with the virtual cranial surface model. The virtual head model and the real participant's head share the same coordinate system S_{head}. The following spatial transformation is computed to convert coordinates planned in the rendering software into applicator positions:

$$T_{render}^{applicator} = T_{robot}^{applicator} T_{cam}^{robot} T_{head}^{cam} T_{render}^{head}$$

A key source of systematic positioning error arises from discrepancies in landmarks' positions captured by the stereo vision camera versus the rendering software camera, even though both the virtual head model and real head use identical facial landmarks to construct S_{head}. MediaPipe is employed to capture 2D facial landmarks, which are converted into corresponding 3D coordinates within both the

stereo camera space and the rendering software space. We then optimize the registration between camera space and rendering software space by fine-tuning three landmarks $X' \in \mathbb{R}^{3\times3}$ of the virtual head model under S_{render}(Fig. 4b), expressed as follows:

$$\min_{X'} \mathcal{L}(X') = \| X' - X_{face} \|_F^2 + \lambda_{target} \| P_{model} - P_{face} \|_2^2 + \lambda_{dis} \| X' - X_{init} \|_F^2$$

where $\|\cdot\|_F$ denotes the Frobenius norm. X_{face} represents the three landmarks which is transformed from S_{cam} to S_{render} via the transformation $\left(T_{render}^{head}\right)^{-1}(T_{head}^{cam})^{-1}$. P_{model} and P_{face} respectively denote the coordinates of the scalp targets in the virtual head coordinate system and the real head coordinate system. X_{init} represents the initial coordinates of the three landmarks on the virtual head model. λ_{target} and λ_{dis} are weight coefficients. This function comprehensively measures the alignment accuracy of facial landmarks between the two spaces, the consistency of scalp targets, and regularization. Numerical solution is performed using the Nelder–Mead simplex method, which does not require an analytical gradient.

Robotic control and safety

The markerless navigation system utilizes a UR5e robot, a 6-DOF robotic arm with a repeatability of 0.1 mm and a maximum payload of 5 kg. The robot is equipped with an integrated force/torque sensor at the end effector. Real-time monitoring is performed via a port that transmits the status of the robotic arm at a 500 Hz frequency. Conventional robotic arm commands, such as motion commands, are transmitted via a secondary port that receives commands at 10 Hz. The compliance control strategy based on admittance control is employed to ensure operational safety, particularly during contact between the end effector and the participant's head. The Z-axis in the $S_{applicator}$ is designated as the sole compliant axis, applied with a constant target force of 4 N. This enables the robot to dynamically adjust its displacement along the contact direction based on real-time force feedback, maintaining stable contact force while preventing excessive pressure on the participant's head. Additionally, speed limits are set on the robot to prevent unintended abrupt movements.

In vivo studies on insomnia

Facial point cloud data were acquired from each of the 18 patients to generate cranial surface models and plan targeting positioning. Patients received rTMS treatment five times per week over a two-week period using the workflow. The right dorsolateral prefrontal cortex (R-DLPFC; MNI: x=44, y=38, z=34) was selected as the treatment target[5] and the MagTD 80 TMS device was employed for treatment. The stimulation parameters for each session were set as follows: a frequency of 1 Hz, an intensity of 80% of the resting motor threshold (rMT), 1200 pulses, with a 1-second interval after every 20 pulses, duration 21 minutes. Clinical efficacy was assessed pre- and post-treatment using the Pittsburgh Sleep Quality Index (PSQI) and Insomnia Severity Index (ISI). 3D-T1W images were acquired from each patient for *in silico* comparison of systematic errors.

Supplementary Information

Supplementary Figs. S1-S3

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Acknowledgments:

Funding: The authors would like to acknowledge the supports of the National Key Research and Development Program of China (2023YFC3603800,2023YFC3603804,2023YFC3603803), the GuangDong Basic and Applied Basic Research Foundation (2025B0303000013).

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Declaration of interests:

Authors declare that they have no competing interests.

Data availability:

Data relating to the face-to-scalp mapping contains facial information and is not made publicly available due to privacy considerations and ethical restrictions. The data used for the cortex-to-scalp mapping modelling is sourced from https://fcon_1000.projects.nitrc.org/indi/retro/southwestuni_qiu_index.html.

Code availability:

The code will be made publicly available at https://github.com/Jerry-brown/Navigated_non-invasive_neuromodulation upon acceptance of this manuscript.